

Assignment: Number Range Validator

Objective: To demonstrate understanding of decision-making using `if/else` statements and loops in Java.

Problem Statement: Create a program that asks the user to input a number within a specified range (1 to 50). The program will verify if the number is within the range and, if not, prompt the user to input a valid number. The program will continue until a valid number is entered, and it will display a message when the input is correct.

Requirements:

1. **User Input:** The program will prompt the user to enter a number between 1 and 50.
2. **Decision-making:**
 - If the number is **less than 1**, inform the user that the number is too low and ask for another input.
 - If the number is **greater than 50**, inform the user that the number is too high and ask for another input.
 - If the number is within the correct range, inform the user that the input is valid.
3. **Looping:** The program must continue asking the user for input until they enter a number between 1 and 50.

Expected Output Example:

Enter a number between 1 and 50: -3
The number is too low. Try again.

Enter a number between 1 and 50: 75
The number is too high. Try again.

Enter a number between 1 and 50: 25
Thank you! 25 is within the range.

Bonus:

1. **Enhanced Feedback:** If the number is out of range, tell the user whether it is too low or too high and also indicate how far they are from the valid range (e.g., "Your guess of 75 is 25 above the upper limit").
2. **Allow Range Customization:** Ask the user to set the minimum and maximum limits for the range before starting the validation.

Starter Code Example:

```
import java.util.Scanner;

public class NumberRangeValidator {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int number = 0;

        // Continue to ask for input until the number is in the range
        while (number < 1 || number > 50) {
            System.out.print("Enter a number between 1 and 50: ");
        }
    }
}
```

```

        number = scanner.nextInt();

        if (number < 1) {
            System.out.println("The number is too low. Try again.");
        } else if (number > 50) {
            System.out.println("The number is too high. Try again.");
        } else {
            System.out.println("Thank you! " + number + " is within the range.");
        }
    }

    scanner.close();
}
}

```

Submission: Submit your .java file by the due date. Ensure that the program behaves as specified and that you have tested it with different inputs.